

Subnational GDP for Indonesia: A District-Level Panel, 1996–2025 — Data Guide

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Abstract

We provide a district-level panel of Indonesian regional GDP (Produk Domestik Regional Bruto, PDRB) covering 1996 to 2025 (2024–25 provisional). The panel reports nominal, real (constant-price), and per-capita PDRB, harmonized to 2000-vintage administrative boundaries so that more than 350 districts stay consistently defined across three decades of administrative splits (*pemekaran*). The panel is built from 21 archival BPS (Badan Pusat Statistik, Statistics Indonesia) publications, parsed from PDF; all source files, crosswalks, scripts, and outputs are included for full replication, and the unharmonized (“native”) panels are retained alongside the harmonized ones. We cross-validate against the World Bank’s INDO-DAPOER database. *Contents:* [Quick Start](#) · [Background: Key Concepts](#) · [§1 Introduction](#) · [§2 File Structure](#) · [§3 Source Data](#) · [§4 Pipeline](#) · [§5 Column Definitions](#) · [§6 Working with the Data](#) · [§7 Known Limitations](#) · [§8 DAPOER Cross-Validation](#) · [§9 Corrections and Provenance](#) · [Errata and contact](#)

Quick Start

The dataset ships five panel files:

File	Series	Years
<code>outputs/panel_pdrb_total_2000bounds.csv</code>	total PDRB, harmonized to 2000 boundaries	1996–2025
<code>outputs/panel_pdrb_capita_2000bounds.csv</code>	per-capita PDRB, harmonized	1996–2025
<code>outputs/panel_pdrb_chained_2000bounds.csv</code>	chain-linked real (2010 base), harmonized	2002–2025
<code>outputs/panel_pdrb_total_native.csv</code>	total PDRB, native (unharmonized)	1996–2025
<code>outputs/panel_pdrb_capita_native.csv</code>	per-capita PDRB, native (unharmonized)	1996–2025

The `_2000bounds` files are the recommended default. Use the `_native` files when you need exactly what a BPS publication printed; use the chained file only for cross-base-year growth analysis.

The panel CSVs themselves are plain text and load in any language. Reproducing the panels from source (§4) requires Python 3.11+; see `requirements.txt` for package versions.

Load a panel and filter to a working series:

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```
import pandas as pd
df = pd.read_csv("outputs/panel_pdrb_total_2000bounds.csv")

# Nominal PDRB, confirmed figures, no pre-split aggregate rows
nom = df[
    (df.table_type == "nominal") &
    (df.is_provisional == False) &
    (df.is_pre_split_aggregate == False)
]
```

Values in `value_standardized` are in rupiah. Divide by `1e6` to match the DAPOER convention (IDR million). Always drop `is_pre_split_aggregate == True` rows before any aggregation; they repeat a child district's value inside its parent and double-count on summation (§7.9).

The three real GDP series are not comparable across base years: `real/1993` (1996–2003), `real/2000` (2002–2012), and `real/2010` (2011–2025) use different price bases and SNA vintages. Jumps at 2003 or 2013 are methodological breaks, not economic signals. **For a continuous constant-price series use the chained panel (§7.8). The nominal series is continuous within each base-year/SNA era but carries a large methodological break at 2011** — when BPS moved from the 2000-base/1993-SNA framework to the 2010-base/2008-SNA framework, nominal PDRB steps up system-wide (median district 2010→2011 ratio ≈ 1.65 vs ≈ 1.13 at ordinary year boundaries). Do not compute 2010↔2011 nominal growth from this panel; see §7.16. A smaller elevation appears at the 2001→2002 boundary (median ≈ 1.21).

Aceh district values for 2000–2001 reflect contemporaneous conflict-period reporting and are substantially below BPS's later retrospective revisions for several districts. For pre-2002 Aceh, restrict to 2002 onward or substitute DAPOER's revised values, documenting the choice (§7.2).

The per-capita `_2000bounds` panel is not territory-consistent across split years. After a district splits, the parent's per-capita series covers the rump territory only, not the full 2000-vintage area. The most visible case is Sumbawa, which drops 66% in 2004 when its mine separates. For a boundary-consistent per-capita series across a split year, divide total-panel PDRB by an external population series (§7.14).

The chained panel's levels are synthetic and not additive across districts. Do not sum them into a province total; use the chained file for growth rates only (§7.8).

Known errors have been corrected and documented (144 corrections; §9), but undetected errors may remain. For analysis sensitive to individual cell values, verify against `parsed_csvs/` (replication package) or the source PDF (§7.15).

Background: Key Concepts

PDRB (Produk Domestik Regional Bruto) is the regional analogue of GDP: the gross value added produced within a district (*kabupaten*/regency or *kota*/city) in a given year. BPS publishes it both at current prices (nominal) and at constant prices of a fixed base year (real). Three accounting regimes appear in this dataset: SNA 1993 at base year 1993 (publications covering 1996–2003), SNA 1993 at base year 2000 (2002–2012), and SNA 2008 at base year 2010 (2011–2025). A base year fixes the price vector used to value real output; an SNA vintage fixes the accounting methodology, including sector definitions, imputations, and coverage. Real levels are therefore not comparable across

regimes: a jump at a rebasing boundary (2003 or 2013) reflects a methodological change rather than an economic event. Nominal values are continuous across the full period.

Since decentralization began in 1999, Indonesia has repeatedly split districts and provinces (*pemekaran*), creating hundreds of new *kabupaten* and *kota*; a district reported as one unit in 1998 may be four units by 2015. To build a consistent panel, we fix a single boundary vintage of 2000 and aggregate later child districts back to their 2000-vintage parents ([Section 4](#) residual complications in [Section 7](#)). BPS also routinely publishes two PDRB variants: a total that includes oil and gas, and a *tanpa migas* variant that excludes the oil-and-gas sector, whose output is volatile and geographically concentrated. Where BPS published both variants, we retain them as separate panel rows. The oil-excluded variant exists only in the publications covering 1996 to 2010.

Sumbawa, in Nusa Tenggara Barat, illustrates the mechanism. In the 2000-vintage map it is a single *kabupaten*, but in 2003 a new *kabupaten*, Sumbawa Barat, split off from it under *pemekaran*. In the `_native` panels the two appear exactly as BPS published them: one Sumbawa row before 2003, then two rows — Sumbawa and Sumbawa Barat — from 2004 onward. In the harmonized `_2000bounds` panel they are instead summed back into a single Sumbawa row for every year, so the series refers to the same 2000-vintage territory throughout. This re-aggregation, applied to roughly 120 post-2000 splits, is 2000-vintage boundary harmonization.

We also provide a chain-linked panel that splices the older series onto the 2010-price base at a 2011 pivot (`table_type == 'real_chained'`). Its levels are synthetic constant-2010-price figures. They are suitable for growth analysis, but they are not additive across districts ([Section 7.8](#)). A small set of observations was manually corrected for confirmed BPS source errors, transcription artifacts, or values recovered from BPS Query Builder exports; every corrected value carries a stable `correction_id` linked to its justification ([Section 9](#)).

1. Introduction

These data come from an undergraduate research project conducted in 2025–2026. Nayfa Johan, a native speaker of Bahasa Indonesia, conducted the primary research; Dr. Russell Hillberry supervised.

The challenge is geographic consistency: if district boundaries change mid-panel, a measured GDP drop may reflect administrative separation rather than economic decline.

INDO-DAPOER covers 2000–2020 and includes no per-capita series. This panel is built from BPS archival publications and documents how every administrative boundary change and methodology break is handled. The pipeline reproduces from the included parsed CSVs (archival PDFs also provided); DAPOER serves as a cross-validation check, not a source.

The unharmonized (native) panels are retained alongside the harmonized ones, so a user can reconstruct exactly what any individual publication printed; every value carries its source publication, a priority rank, its harmonization status, and any correction identifier. Where two publications report the same district-year, we use the later one — BPS revisions reflect improved field coverage — with the earlier figure visible in the native panels. Known source errors, boundary seams, and structural gaps are documented in [Section 7](#) rather than quietly patched.

2. File Structure

The GitHub repository is sufficient on its own to reproduce every published panel: `bash run_all.sh` runs Stages 2–7 straight through, starting from the tracked parsed CSVs (Stage 1's output — see `pipeline_out/` below), and reproduces the five output files byte-for-byte. The raw source PDFs themselves are not tracked in this repository; they're needed only if you want to verify a specific transcribed value against the original scanned or digital page BPS printed, one level deeper than reproducing the panel. For that, they ship in the PURR replication deposit (DOI 10.4231/FWEQ-VE94; full citation in ["Errata and contact"](#) below), alongside everything else here.

The project root contains:

Directory	Contents
<code>raw/</code>	Source material: <code>raw/pdf/</code> holds the BPS publication PDFs; <code>raw/csv/</code> and <code>raw/parsed/</code> hold intermediate parse artifacts. Ships in the PURR replication deposit only — not tracked in this repository (see above).
<code>parsed_csvs/</code>	The 21 parsed CSVs (Stage 1 output), verbatim transcriptions of the source PDFs — the reproducibility anchor for the whole pipeline. Tracked in this repository at <code>pipeline_out/full_runs/csvs/</code> ; the replication package presents the same files flattened at <code>parsed_csvs/</code> . Stage 2 reads these and writes corrected copies to <code>corrected_csvs/</code> , which is regenerated at runtime and not shipped.
<code>corrections/</code>	<code>corrections/corrections_applied.csv</code> , the correction sidecar. In the replication package. (The external CSVs used for recoveries are in the full project repository, not the replication package; see Section 9 .)
<code>pipeline_out/</code>	Intermediate pipeline outputs: <code>pipeline_out/standardized/</code> = Stage 3, <code>pipeline_out/harmonized/</code> = Stage 4, <code>pipeline_out/audits/</code> = Stage 6 QA reports (all regenerated at runtime, not tracked, not in the replication package); <code>pipeline_out/full_runs/csvs/</code> = the tracked parsed CSVs (see <code>parsed_csvs/</code> above).
<code>outputs/</code>	The published panels (see inventory below). These are the files a data user reads.
<code>crosswalks/</code>	Harmonization inputs and supporting pemekaran (administrative-split) reference material — 8 files, itemized in “Crosswalk file reference” below.
<code>dapoer_extract/</code>	The World Bank INDO-DAPOER extract used for cross-validation (Section 8): <code>dee68668-..._Data.csv</code> (the indicator values) and <code>dee68668-..._Series - Metadata.csv</code> (indicator/topic definitions), both in the World Bank Data Bank export format. External files, not authored by this project — see “DAPOER extract file reference” in Section 8 .
<code>scripts/</code>	The pipeline scripts (Section 4) plus one-off fix scripts and audit utilities.
<code>docs/</code>	This data guide (<code>data_guide.md</code>) and the methodology and decision log (<code>research_log.md</code>). The data guide ships in the replication package; <code>research_log.md</code> is in the full project repository only.

Crosswalk file reference

All 8 files in `crosswalks/`:

File	Rows	Columns	Purpose
<code>district_reference.csv</code>	422	<code>district_id</code> , <code>district_num</code> , <code>province_canonical</code> , <code>district_name_2000</code> , <code>admin_type</code> , <code>bps_code</code> , <code>multi_province_flag</code> , <code>notes</code>	The canonical 2000-vintage district list; the harmonization target every panel row is matched against. 422 rows for 421 districts because one is a deliberate alias (see note below).
<code>province_crosswalk.csv</code>	42	<code>province_name_clean</code> , <code>province_canonical</code> , <code>province_canonical_en</code> , <code>change_type</code> , <code>split_from</code> , <code>split_year</code> , <code>law_number</code> , <code>notes</code>	Maps raw BPS province name strings (across spelling variants and provincial splits) to a single canonical name.
<code>name_corrections.csv</code>	122	<code>region_name_base_raw</code> , <code>region_name_base_corrected</code> , <code>correction_type</code> , <code>source_files</code> , <code>notes</code>	Hand-verified fixes for OCR/typographic district-name errors that would otherwise fail to match <code>district_reference.csv</code> .
<code>split_concordance.csv</code>	172	<code>child_name_raw</code> , <code>child_base</code> , <code>parent_name_raw</code> , <code>parent_base</code> , <code>parent_base_2000</code> , <code>province</code> , <code>year</code> , <code>law_number</code> , <code>date_raw</code> , <code>in_standardized</code> , <code>parent_in_standardized</code> , <code>notes</code>	The <i>pemekaran</i> (administrative-split) parent↔child map: for every post-2000 child district, its 2000-vintage parent, the enabling law, and effective date. The backbone of the <code>_2000bounds</code> harmonization.
<code>bps_code_reference.csv</code>	485	<code>bps_code</code> , <code>bps_name</code> , <code>province</code> , <code>province_code</code> , <code>admin_type</code> , <code>code_vintage</code> , <code>notes</code>	BPS's own Kode Wilayah reference across code vintages (see <code>bps_code</code> caveats in Section 5).
<code>RISED - Data Pemekaran Daerah (Kab_Kota).csv</code>	511	No, Kode Provinsi, Nama Provinsi, Kode Kab./Kota, Nama Daerah, Klasifikasi, Kode Induk, Daerah Induk, Tahun Mekar (UU), Tahun DAU	External reference: RISED's compiled administrative-split registry, used as a cross-check on <code>split_concordance.csv</code> during construction; not read by any pipeline script.
<code>Data-otda_daerah_otonom_pemekaran_19942014_kabkot-...csv</code>	215	<code>id</code> , <code>nama_daerah_otonom</code> , <code>hasil_pemekaran_ibukota</code> , <code>hasil_pemekaran_uu</code> , <code>pembentukan</code> , <code>tentang</code> , <code>tggl_berlaku_uu</code> , <code>daerah_induk</code> , <code>wilayah_provinsi</code> , and 10 further fields (area, population, and sub-district counts pre/post split)	Kemendagri's (Ministry of Home Affairs) official 1994–2014 <i>otonomi daerah</i> (regional autonomy) split registry, in Indonesian — a second cross-check on <code>split_concordance.csv</code> , also not read programmatically.
<code>Data Pemekaran wilayah administratif di Indonesia... 2022.docx</code>	—	(document, tabular)	A compiled 1999–2022 administrative-split table by province, in Indonesian. Same cross-check role as the two rows above, not a pipeline input either.

On `district_reference.csv`'s 422nd row: `SULAWESI_TENGGARA_KONawe` (district_num 376) appears twice, with `district_name_2000` “KENDARI” on one row and “KONawe” on the other. Kabupaten Kendari was renamed Kabupaten Konawe in 2004 (UU 6/2003), and the “KENDARI” row is a deliberate alias so that 1996–2003 publications, which still print the old name, route to the same `district_id` as the post-2004 “KONawe” rows. Keying on `district_id` (not `district_name_2000`) avoids any ambiguity.

Published panel inventory (outputs/)

Table 1. Published panels (produced outputs). The five output panels (contrast Tables 2–3, the BPS input sources).

File	Boundaries	Series	Rows	Districts	Years
<code>panel_pdrb_total_2000bounds.csv</code>	2000-vintage (harmonized)	total PDRB	23,742	361	1996–2025
<code>panel_pdrb_total_native.csv</code>	native (unharmonized)	total PDRB	31,034	all incl. post-2000 children	1996–2025
<code>panel_pdrb_capita_2000bounds.csv</code>	2000-vintage (harmonized)	per-capita PDRB	10,400	363	1996–2025
<code>panel_pdrb_capita_native.csv</code>	native (unharmonized)	per-capita PDRB	13,894	all incl. post-2000 children	1996–2025
<code>panel_pdrb_chained_2000bounds.csv</code>	2000-vintage (harmonized)	chain-linked real (2010 base)	8,315	348	2002–2025

The `_2000bounds` files are the recommended default. Use the `_native` files when you need exactly what a publication printed; use the chained file only for cross-base-year real growth analysis, subject to [Section 6](#) and [Section 7.8](#). (2024–25 provisional as of deposit, Aug 2026.) Intermediate outputs (`pipeline_out/`) are generated locally by `run_all.sh` and are not shipped.

3. Source Data

Raw PDF provenance and republication

`raw/pdf/originals/` (replication package) holds 11 curated extracts from the BPS publications: each source publication runs to several hundred pages across many statistical topics, so what’s shipped is the PDRB-relevant page range pulled from each one, not the complete multi-topic volume. Alongside them sits `Term of Use - BPS-Statistics Indonesia.pdf`, a saved copy of BPS’s Terms of Use page (bps.go.id/en/term-of-use, accessed September 2026) kept as dated documentary evidence rather than a link that could change. Clause 13 of that page grants content “free of charge, worldwide, on a continuous and non-exclusive basis” for, among other purposes, “using the data for both commercial and non-commercial purposes” and “copying, distributing, and/or transmitting the content,” conditional on lawful use, proper citation (title, access date, and a link to the original — see footnote 1 below and the

publication table in this section), and accepting that content may change or be withdrawn. These pages are shared on that basis.

They are the extracted pages, not touched-up copies — no re-typesetting, no OCR cleanup, no accessibility remediation. Most were produced as page-range exports (macOS Preview’s PDF export; several carry a “Quartz PDFContext” producer tag as a result), so they aren’t byte-identical to BPS’s own files, but they are visually and textually faithful to the pages as BPS printed them, which is what verifying a panel figure against source actually requires. Accessibility and general usability live in the derived data instead: `outputs/` and `parsed_csvs/` (“Parsed CSV schema” below) are plain, machine-readable text.

The panel is built from 21 BPS source files²: 11 total-PDRB source files and 10 per-capita source files. (The 11 total sources include the `PDRB_1998-2001_jabar_banten` file, a supplementary BPS volume added to fill the Banten and post-split Jawa Barat 2000–2001 window; see [Section 7.7](#).) Source files correspond one-to-one with the parsed CSVs in `parsed_csvs/` (replication package). BPS releases rolling multi-year windows; overlaps are resolved by source priority ([Section 4](#), Stage 5).

Total-PDRB source files

Table 2. Total-PDRB source publications (BPS input data). Archival BPS volumes ingested by the pipeline. Each row is one published volume; base year and SNA vintage fix the price and methodology regime (see [Background](#)). Windows overlap by design and are reconciled by source priority in Stage 5.

²BPS publication portal: <https://www.bps.go.id/id/publication?keyword=Produk+Domestik+Regional+Bruto+Kabupaten%2FKota+di+Indonesia+2020-2024&sort=latest> (search interface; accessed 14 August 2026).

Source file (window)	Base year	SNA vintage	Parsing notes
PDRB_1996-1999	1993	SNA 1993	Physical scan PDF; parsed in vision mode
PDRB_1998-2001_jabar_banten	1993	SNA 1993	Supplementary BPS volume; supplies Banten + post-split Jawa Barat 2000–2001 (Section 7.7)
PDRB_2000-2001	1993	SNA 1993	Digital PDF; constant-price tables are still base-1993 despite the publication title (see Coverage Summary below)
PDRB_2002-2004	2000	SNA 1993	Digital PDF
PDRB_2005-2007	2000	SNA 1993	Image-only PDF; parsed in vision mode
PDRB_2008-2010	2000	SNA 1993	Digital PDF
PDRB_2011-2013	2010	SNA 2008	Digital PDF
PDRB_2014-2016	2010	SNA 2008	Digital PDF
PDRB_2017-2019	2010	SNA 2008	Digital PDF
PDRB_2020-2022	2010	SNA 2008	Digital PDF
PDRB_2021-2023	2010	SNA 2008	Extends provisionally to 2025

Per-capita source files

Table 3. Per-capita PDRB source publications (BPS input data). Ten per-capita volumes parallel the ten main total-PDRB volumes, one per window. (The per-capita panel is nominal.)

Source file (window)	Base year	SNA vintage	Parsing notes
PDRB-capita_1996-1999	1993	SNA 1993	Physical scan PDF; parsed in vision mode
PDRB-capita_2000-2001	2000	SNA 1993	Sumatera Barat 2000–2001 corrected using BPS Query Builder (Section 9, W2c). Per-capita is nominal only; base year reflects the publication label (contrast PDRB_2000-2001 in Table 2, which is base-1993 in content despite its title).
PDRB-capita_2002-2004	2000	SNA 1993	Digital PDF
PDRB-capita_2005-2007	2000	SNA 1993	Image-only PDF; parsed in vision mode
PDRB-capita_2008-2010	2000	SNA 1993	Digital PDF
PDRB-capita_2011-2013	2010	SNA 2008	Kalimantan Utara column shift corrected (Section 9, KALTARA_shift)
PDRB-capita_2014-2016	2010	SNA 2008	Digital PDF
PDRB-capita_2017-2019	2010	SNA 2008	Five Lampung districts had wrong population denominators; corrected (Section 9, W2/W2b)
PDRB-capita_2020-2022	2010	SNA 2008	Digital PDF
PDRB-capita_2021-2023	2010	SNA 2008	Extends provisionally to 2025

The per-capita panel includes 1996–1999 data for 139 districts from the 1996–1999 per-capita volume, but several districts (including Banda Aceh and Ambon) lack pre-2000 per-capita observations; their coverage begins in 2000.

Coverage summary

Panel	Districts	Years	Rows
total_2000bounds	361	1996–2025	23,742
capita_2000bounds	363	1996–2025	10,400
chained	348	2002–2025	8,315

Real-series windows in the (non-chained) total panel: - **real/1993**: 1996–2003 (SNA 1993, base 1993) - **real/2000**: 2002–2012 (SNA 1993, base 2000; the 2000–2001 publication’s constant-price

tables are still base-1993) - **real/2010**: 2011–2025 (SNA 2008, base 2010) - **nominal**: continuous 1996–2025 - **oil-excluded**: available in 1996–2010-era publications only.

Parsing methodology (including LLM use)

We parsed the BPS PDF tables to wide-format CSVs with a large language model (Claude), which transcribes each table verbatim, preserving Indonesian number formatting. Two publications are physical scans (1996–1999 total and 2005–2007 total) and were parsed in vision mode, in which the model reads the rendered page image rather than an embedded text layer. Numeric interpretation (format conversion, series-type extraction, unit handling) happens at later deterministic stages, so parse output can be diffed directly against the source page. Every parsed CSV is retained in `parsed_csvs/` (replication package).

LLM transcription introduces occasional character-level errors, and the BPS sources themselves contain errors; a manual correction stage applies 144 content-keyed corrections (22 fix families), each documented and reversible (Section 9).

Parsed CSV schema (`parsed_csvs/`)

Every file in `parsed_csvs/` shares this raw, pre-standardization schema:

Column	Definition
<code>province_name</code>	Province name exactly as printed on the page (not yet harmonized).
<code>regency_code</code>	The district’s BPS code number as printed in the table’s leftmost column.
<code>region_name_raw</code>	District name exactly as printed, including OCR artifacts.
<code>name_flag</code>	Footnote marker attached to the name (e.g. 1) for an oil-excluded row), where the source prints one.
<code>year</code>	Column year, taken from the table header.
<code>year_flag</code>	Set if the year required disambiguation (e.g. an ambiguous or malformed header).
<code>value</code>	The printed number, transcribed verbatim in its original Indonesian formatting (before <code>unit_multiplier</code> is applied).
<code>value_flag</code>	Set on a value the parser could not confidently transcribe (e.g. an OCR garble); these are the candidates the correction pass (Section 9) resolves.
<code>table_header_raw</code>	The table’s Indonesian-language title, transcribed verbatim.
<code>table_header_english</code>	The table’s English-language title, transcribed verbatim. A small number of these carry OCR typos from the source PDF’s own printed English caption (e.g. “Municpalities” for “Municipalities” in <code>PDRB_2000-2001.csv</code> ’s DKI Jakarta table). The field is descriptive only — free text, not used in any parsing, matching, or join logic — so the typo doesn’t reach the pipeline or the published values, and it’s outside the correction sidecar’s scope for that reason.
<code>page_number</code>	Source PDF page the row was read from.
<code>table_number</code>	Sequential table index within the source file, used with <code>page_number</code> to locate a row exactly.
<code>raw_text_hash</code>	Content hash of the row, used downstream to key corrections (Section 9) and to detect duplicate rows (below).

Duplicate rows from resumed parsing runs. Six files in the replication package contain literal, byte-for-byte duplicate rows — same page, same value, same `raw_text_hash`, appearing twice: `PDRB-capita_1996-1999.csv` (648 of 1,116 rows), `PDRB_1996-1999.csv` (960 of 4,036), `PDRB_2000-2001.csv` (46 of 4,384), `corrections/corrections_applied.csv` (36 of 166), and the two `dapoer_extract/` files (3 rows each, not further diagnosed — external World Bank export). The cause is specific pages being re-parsed on a resumed run and appended rather than replacing the original parse. This is a known characteristic of the parsing stage — `standardize_pipeline.py` explicitly deduplicates on `raw_text_hash` before building the panel (see its Stage 3 comment), which is why the published panels and `crosswalks/` files carry zero duplicate rows regardless. The one file where this dedup is **not** applied downstream is `corrections/corrections_applied.csv`, so its 36 duplicate rows are visible as shipped; they are inert (each pair records the same correction twice, not two different corrections) but a straight row count of that file will overstate the correction count by exactly its duplicate rows. Use the sidecar's distinct-row count for any exact reconciliation (Section 9 gives the distinct counts already).

4. Pipeline

The pipeline runs in seven stages. Scripts live in `scripts/`. Stage 1 requires a paid vision API and is not re-run in replication; its output ships in `parsed_csvs/`. `run_all.sh` chains Stages 2–7 in order and writes SHA-256 checksums of the five output panels.

Stage 1: Parse. `parse_pdfs_text.py` / `parse_pdfs_scan.py` / `reparse_pages_vision.py` -> `parsed_csvs/` Emits one wide-format CSV per publication, transcribed verbatim. `parse_pdfs_text.py` processes the digital PDFs (2000–2023) in text-extraction mode; `parse_pdfs_scan.py` handles the 1996–1999 physical scan in vision mode. `reparse_pages_vision.py` re-parses individual pages in vision mode where text extraction misassigns column positions (sparse new-district rows and image-only pages).

Stage 2: Corrections. `apply_corrections.py` -> `corrected_csvs/` (**regenerated at runtime, not shipped**) Applies the 144 corrections for confirmed BPS source errors, transcription artifacts, and BPS Query Builder recoveries; the source files are not modified. Each correction is content-keyed, matched on (province, district, year, current value) rather than row index, so it survives re-parsing and re-ordering. Each resolves to one of three states: apply (the bad value is found and corrected), skip (the good value is already present — the correction was previously applied), or hard-fail (the value is neither the expected bad value nor the corrected one, which signals an unexpected upstream change and stops the run for inspection). A sidecar, `corrections/corrections_applied.csv`, records every applied correction with its `correction_id` (Section 9).

Stage 3: Standardize. `standardize_pipeline.py` -> `pipeline_out/standardized/` (**21 files**) Converts Indonesian-formatted strings to floats and extracts series metadata from headers and row structure: `table_type`, `base_year`, `oil_excluded`, `admin_type`, and `unit/unit_multiplier`. Deterministic rules handle known quirks: a recurring garbled header, BPS structural sub-total markers that mimic oil-excluded rows, and a unit override for one mislabeled Aceh per-capita table.

Stage 4: Harmonize. `harmonize_regions.py` -> `pipeline_out/harmonized/` (**21 files**) Assigns the canonical `district_id`, applies name corrections (`crosswalks/name_corrections.csv`), and resolves pemekaran via `crosswalks/split_concordance.csv`; the legal basis (UU number) for

each split was verified against JDIH (peraturan.bpk.go.id) and Kemendagri pemekaran records.³ Post-2000 child districts are aggregated back to their 2000-vintage parent (`is_child_aggregate = True` on summed rows), subject to an early-data carve-out: a concordance child that appears as a standalone, non-#) data row in an early file (1996–1999 or 2000–2001) is treated as its own 2000-vintage unit, regardless of magnitude. Every row receives a `match_status`.

Stage 5: Build panel. `build_panel.py` -> `outputs/` Stacks the 21 harmonized files and enforces uniqueness on the panel key (`district_id, year, table_type, base_year, oil_excluded`). On a collision, the higher `source_priority` (later publication) wins, and non-provisional rows are promoted over provisional ones.

Stage 6: Validate. `validate_panel.py` -> `pipeline_out/audits/` Eleven QA checks: duplicate keys, within-district year gaps, growth-rate anomalies, province-sum reconciliation, implied-population and implied-deflator smoothness, and roster completeness across publications. Reports are reviewed manually and inform [Section 7](#).

Stage 7 (optional): Chain-link. `chain_link.py` -> `outputs/panel_pdrb_chained_2000bounds.csv` Splices the real/2000 series onto the real/2010 base using `link_year = 2011`, producing a synthetic constant-2010-price level series. See [Section 7.8](#).

5. Column Definitions

Columns are defined for the primary panels (`panel_pdrb_total_2000bounds.csv`, `panel_pdrb_capita_2000bounds.csv`), which share an identical column set. Columns present only on the chained panel are marked **(chained only)** columns present on the `_native` panels and the chained panel, but not `_2000bounds`, are marked **(native + chained only)**.

³Kemendagri PELITA: <https://pelita.kemendagri.go.id/>. Kemendagri daerah otonom pemekaran data (kabupaten/kota, 1994–2014): <https://ppid.kemendagri.go.id/front/dokumen/detail/500391609>. RISED Pemekaran Daerah Kabupaten/Kota dataset: <https://figshare.com/> (search “RISED Pemekaran”).

Identification

Column	Type	Definition
<code>district_id</code>	string	Canonical permanent identifier at 2000-vintage boundary, format <code>PROVINCE_DISTRICT_NAME</code> (e.g., <code>ACEH_BANDA_ACEH</code>). Where a kabupaten and kota share the same name, a <code>_KAB</code> or <code>_KOTA</code> suffix disambiguates (e.g., <code>JAWA_BARAT_BOGOR_KAB</code> , <code>JAWA_BARAT_BOGOR_KOTA</code>). Stable across the panel.
<code>year</code>	integer	Calendar year of observation.
<code>district_num</code>	integer	District sequence number in the BPS reference list (<code>crosswalks/district_reference.csv</code>).
<code>bps_code</code>	string	Official BPS 4-digit Kode Wilayah (e.g., 1107 for Aceh Barat). Standard cross-dataset join key for BPS administrative data. Two caveats: (1) roughly 15 pre-split parent/successor pairs share the same code (e.g., <code>BUNGOTEBO</code> and <code>BUNGO</code> both map to 1509), so <code>bps_code</code> alone is not a unique key; always join on <code>district_id</code> ; (2) the province prefix of <code>bps_code</code> does not match <code>province_canonical</code> for three province-migrated districts (<code>POLEWALIMAMASA</code> , <code>PASANGKAYU</code> , <code>KEPULAUAN_RIAU</code>).
<code>province_canonical</code>	string	Harmonized province name, all-caps.
<code>district_name_2000</code>	string	Canonical district name at 2000-vintage boundary, all-caps.

The four columns below are carried only on the `_native` panels and the chained panel, not `_2000bounds` (see the marking convention above):

Column	Type	Definition
<code>admin_type</code>	string	<code>kabupaten</code> (regency) or <code>kota</code> (city). Assigned from the BPS regency code where the publication did not label it: <code>kota</code> if <code>(bps_code mod 100) ≥ 71</code> , else <code>kabupaten</code> . Used internally during harmonization for all panels to disambiguate kab/kota pairs; retained as an output column on the <code>_native</code> and chained panels only, not <code>_2000bounds</code> .
<code>region_name_corrected</code>	string	The parsed district name (<code>region_name_base</code>) after applying <code>crosswalks/name_corrections.csv</code> — i.e. with known OCR/typographic errors fixed, but before harmonization to a <code>district_id</code> . Used internally to identify post-2000 child districts by name (they have no <code>district_id</code>); retained as an output column for auditing the harmonization.
<code>is_post2000_child</code>	boolean	<code>True</code> if this row is a post-2000 split (child) district that did not exist as a distinct BPS entry before 2000 — e.g. Kota Cimahi. <code>False</code> for districts that stood as their own BPS entry in the 2000-vintage boundary set. Used internally to route rows to child-summation in <code>_2000bounds</code> ; retained as an output column on <code>_native</code> and chained.
<code>parent_district_id</code>	string / NaN	For rows where <code>is_post2000_child == True</code> : the <code>district_id</code> of the 2000-vintage parent this child sums into for the <code>_2000bounds</code> panels. <code>NaN</code> if the parent could not be resolved, or if the row is not a post-2000 child.

Series key

Column	Type	Definition
<code>table_type</code>	string	nominal (current price), <code>real</code> (constant price), or <code>real_chained</code> (chained panel only).
<code>base_year</code>	integer / NaN	Base year of a real series: 1993, 2000, or 2010. NaN for nominal. The three base years are not comparable across breaks.
<code>oil_excluded</code>	boolean	<code>True</code> = PDRB <i>tanpa migas</i> (excluding oil and gas). <code>False</code> = total PDRB (oil-inclusive). Both variants are separate panel rows where published. Oil-excluded rows exist only for 1996–2010-era publications; for 2011+ rows, <code>oil_excluded == False</code> means the row is the oil-inclusive total — those publications carry no oil-excluded variant.

Together with `district_id` and `year`, `(table_type, base_year, oil_excluded)` form the enforced unique panel key.

Value

Column	Type	Definition
<code>value_standardized</code>	float	PDRB value in rupiah (total panel) or rupiah per person (per-capita panel). BPS reported in mixed units (jutaan/miliar/ribu rupiah); all have been converted to rupiah by applying <code>unit_multiplier</code> . No further multiplier is needed at read time.
<code>unit</code>	string	Original BPS unit string (e.g., <code>juta rupiah</code> , <code>miliar rupiah</code> , <code>ribu rupiah</code> , <code>rupiah</code>).
<code>unit_multiplier</code>	float	Factor applied to reach rupiah: <code>1e6</code> (juta), <code>1e9</code> (miliar), <code>1e3</code> (ribu), <code>1.0</code> (rupiah).

Provenance

Column	Type	Definition
<code>source_file</code>	string	BPS publication CSV filename (e.g., <code>PDRB_2021-2023.csv</code> , <code>PDRB-capita_1996-1999.csv</code>).
<code>source_priority</code>	integer (0–10)	Publication rank; higher = later. On a key collision, the higher-priority row wins (“later wins”), reflecting BPS field-coverage revisions.
<code>match_status</code>	string	Harmonization outcome (Stage 4). Enumerated values below.
<code>correction_id</code>	string / NaN	Fix identifier linking this row to the correction sidecar (§9). Present on <code>_native</code> and chained panels; NaN if the row was not corrected. Not present on <code>_2000bounds</code> panels (child summation collapses multiple source rows into one).

The `match_status` values are:

Value	Meaning
<code>matched</code>	Direct match on <code>(province, district_name_2000)</code> .
<code>matched_with_admin_type</code>	A district whose name is shared by a <i>kabupaten</i> and a <i>kota</i> (e.g. Bogor, Bandung, Serang), disambiguated to the correct unit using the source <code>admin_type</code> . This covers 3,389 rows in the total <code>_2000bounds</code> panel and 1,500 in the per-capita panel. One historical rename means <code>district_reference.csv</code> carries 422 rows for 421 2000-vintage districts; analysts who join on that file should key on <code>district_id</code> , not name. See the research log for the specific case.
<code>matched_province_fallback</code>	District matched by name alone; province inferred from adjacent rows, a lower-confidence provenance (231 rows in the total <code>_2000bounds</code> panel, 54 in the per-capita <code>_2000bounds</code> panel).
<code>child</code>	Harmonize-layer: a post-2000 split (child) district passed through to the full granular panel with <code>district_id</code> null. Present only in the <code>_native</code> panels.
<code>child_aggregate</code>	Build-panel-layer: set on standalone orphan rows produced during child summation when the 2000-vintage parent is absent from that source file (the <code>_2000bounds</code> total panel). Distinct from <code>is_child_aggregate</code> , which flags the summed-parent rows. Correct at child granularity; incomplete for harmonized-parent totals, because the rump parent was absent from the source.

`correction_id` is a separate column; it does not appear as a `match_status` value. The `_2000bounds` panels do not carry it: child summation collapses multiple source rows into one. To trace corrections for the `_2000bounds` panels, use the sidecar directly ([Section 9](#)).

Flags

Column	Type	Definition
<code>is_provisional</code>	boolean	<code>True</code> if BPS marked the year preliminary (<i>angka sementara / angka sangat sementara</i>). Provisional rows are kept; source-priority dedup promotes non-provisional over provisional for the same key.
<code>is_pre_split_aggregate</code>	boolean	<code>True</code> if BPS printed a “pre-split aggregate” sub-row (marked #): a child district shown before its formal split, whose value double-counts into the parent. The #) marker is structural, not a migas indicator, so these rows are not treated as oil-excluded. Filter these out before any aggregation.
<code>is_child_aggregate</code>	boolean	<code>True</code> if the row was produced by summing post-2000 child districts back to their 2000-vintage parent (the pemekaran aggregation). <code>False</code> if the district stood alone in the publication. Total panel only; always <code>False</code> in the per-capita panel, whose split parents continue as rump-territory series (Section 7.14).
<code>is_backcast (chained only)</code>	boolean	<code>True</code> if the observation was backcast from the 2010 base using the real/2000 growth rate. Backcast values are synthetic 2010-price levels, not BPS-published.
<code>link_year (chained only)</code>	integer	Pivot year used for chain-linking (2011 throughout).
<code>link_ratio (chained only)</code>	float	Ratio of real/2010 to real/2000 at the link year, used to rescale pre-2011 real/2000 values onto the 2010 base.
<code>ratio_drift_flag (chained only)</code>	boolean	<code>True</code> if the district’s real/2000 and real/2010 growth rates disagree by >5% at the 2012 validation year (one year past the pivot). 10 districts are flagged; treat their backcasts with extra caution.

6. Working with the Data

The panel CSVs load directly in Stata, R, Python, or Excel without transformation.

```
import delimited "panel_pdrb_total_2000bounds.csv", stringcols(_all) clear
```

```
df <- read.csv("panel_pdrb_total_2000bounds.csv", stringsAsFactors = FALSE)
```

```
df = pd.read_csv("panel_pdrb_total_2000bounds.csv")
```

Section 5 defines every column; the notes below flag the most common sources of silent errors.

Unit. `value_standardized` is in rupiah — the `unit_multiplier` column has already been applied. Do not multiply again. To convert to IDR millions (the DAPOER convention), divide by 1,000,000.

Booleans. `is_provisional`, `is_pre_split_aggregate`, and `is_child_aggregate` are stored as `True/False` text in the CSV (primary panels); `is_backcast` follows the same convention in the chained panel. pandas and R auto-convert these to native logical types on read, so `== False / == FALSE` works directly. Stata loaded with `stringcols(_all)` keeps them as strings, so filter with `keep if is_pre_split_aggregate == "False"` (quoted).

Nominal rows have no base year. `base_year` is blank for nominal rows — expected, not missing data. Real levels are not comparable across base years (1993, 2000, 2010); always filter to one base year before comparing levels or growth rates.

Panel key. The unique key is (`district_id`, `year`, `table_type`, `base_year`, `oil_excluded`). Verify uniqueness on this key before summing districts or provinces to avoid double-counting pre-split aggregate rows.

Chained panel. The chained panel values are synthetic 2010-price levels and are not additive across districts; use it for growth rates only, never sum to a province total (Section 7.8). Backcast rows (`is_backcast == True`) are model-derived; ten districts carry `ratio_drift_flag == True` and deserve extra caution.

7. Known Limitations

The issues below are features of the data, pipeline decisions, or inherited BPS conventions — not errors.

§	Issue	User action
7.1	Three real series are discontinuous across base-year regimes	Never compare real levels across a publication boundary; use the chained panel for a continuous constant-price series
7.2	Aceh 2000–2001 reflects contemporaneous BPS, not the later revision	Restrict to <code>year >= 2002</code> , or substitute DAPOER values by hand and document it
7.3	Fakfak 1996–1999 nulled (boundary change)	No action — rows already nulled; Fakfak is valid from 2000 on
7.4	MAMASA carried under SULAWESI BARAT even pre-2004	Expect <code>SULAWESI_BARAT_MAMASA</code> throughout; no action on values
7.5	Tambrau output attributed entirely to SORONG	Note MANOKWARI aggregate is slightly understated; adjust manually if material
7.6	Per-capita denominator errors corrected for Lampung + Sumatera Barat only	Verify other provinces, or compute per-capita from total panel ÷ independent population
7.7	Banten + West Java 2000–2001 gap filled from supplementary volume	Do not silently impute residual absences
7.8	Chained series is non-additive; backcasts are synthetic levels	Do not sum districts; use for growth analysis only
7.9	Pre-split aggregate rows double-count	<code>df[df.is_pre_split_aggregate == False]</code> before aggregating
7.10	Papua UU 26/2002 promulgation date inconsistent (Oct vs Dec)	Year is correct; ignore unless exact date matters
7.11	Provisional rows extend to 2025	Filter <code>df.year <= 2023</code> unless provisional years are needed
7.12	Residual year-misassignment risk for sparse split-off rows	Verify against source PDF if exact split-year dynamics matter
7.13	Total and per-capita panels have different district-year coverage	Prefer total panel ÷ independent population for per-capita work
7.14	Per-capita <code>_2000bounds</code> denominator shifts at split years	Do not read per-capita across a split year as one continuous territory
7.15	Undetected errors may remain	Verify in <code>parsed_csvs/</code> and, if needed, the source PDF
7.16	Nominal series carries a 53% system-wide step at the 2011 SNA2008 rebasing	Do not compute 2010→2011 nominal growth; use the chained panel for continuous constant-price levels instead

7.1 The three real GDP series are discontinuous

`real/1993` (1996–2003), `real/2000` (2002–2012), and `real/2010` (2011–2025) correspond to different base years and different SNA vintages (SNA 1993 for the first two, SNA 2008 for the third). Cross-series level comparisons are invalid; apparent change at a publication boundary (2003 or 2013) is a methodological break, not an economic signal. The overlap years (2002–2003 and 2011–2012) allow ratio-based splicing, which the chained panel uses, at the cost of synthetic levels (Section 7.8). The nominal series does **not** avoid this problem the way it might seem to — see Section 7.16 for its own break at 2011.

7.2 Aceh 2000–2001 vintage seam

All Aceh district values for 2000–2001 derive from the 2000–2001 BPS publication and reflect contemporaneous conflict-period reporting. A cross-check against INDO-DAPOER shows substantially higher values for several Aceh districts in this period. DAPOER's 2000–2001 figures are consistent with its own 2002+ series, because DAPOER used the same later BPS retrospective revision that our panel picks up from 2002 onward. Ours are contemporaneous BPS publications; DAPOER's are a later revision. For analysis touching pre-2002 Aceh, choose a vintage explicitly. One option is to restrict the analysis window to 2002 and later (our 2002+ Aceh data agree with DAPOER exactly). The alternative is to substitute DAPOER's 2000–2001 Aceh values manually, since DAPOER district-level PDRB is publicly available from the World Bank⁴; document the substitution.

7.3 Fakfak 1996–1999 is unusable (BPS source error, nulled)

BPS printed nominal Fakfak values of ~4.35T / 4.63T / 12.5T / 11.4T rupiah for 1996–1999, then a ~96% drop to the 2000 value, a ~25× collapse. This reflects an administrative boundary change (old Fakfak encompassed areas now in Sorong and Manokwari), not an economic collapse; the `real/1993` series shows the same discontinuity. All 1996–1999 Fakfak rows are nulled. Fakfak from 2000 onward reflects the smaller post-reorganization boundary and is retained.

7.4 MAMASA cross-province parent seam (resolved; retained in `_2000bounds`)

MAMASA split from POLEWALI MAMASA in 2002, before SULAWESI BARAT split from SULAWESI SELATAN in 2004. MAMASA appears in both `_2000bounds` panels as its own 2000-vintage unit (`SULAWESI_BARAT_MAMASA`), with 48 rows in the total panel and 23 in the per-capita panel (2003–2025). MAMASA is carried under its later (SULAWESI BARAT) provincial assignment throughout, including for years before that province formally existed.

7.5 Tambrauw dual-parent limitation

TAMBRAUW (created by UU 56/2008) split from both Sorong and Manokwari, but the split concordance assigns a single primary parent (SORONG). Any 2000-vintage aggregation therefore attributes all of Tambrauw's output to SORONG's territory, understating MANOKWARI's aggregate. The effect is small, but analysts working at the West Papua provincial level or on either parent should account for it.

⁴World Bank Indonesia Database for Policy and Economic Research (DAPOER): <https://databank.worldbank.org/source/indonesia-database-for-policy-and-economic-research> (accessed 14 August 2026).

7.6 Per-capita wrong-denominator errors: Lampung and Sumatera Barat corrected; other provinces not cross-checked

Two confirmed instances of wrong population denominators in BPS per-capita publications have been corrected:

- **2017–2019 publication, five Lampung districts.** The national publication's per-capita values conflict with those published on BPS Lampung's own provincial website; the provincial figures were adopted as correct (`correction_id = W2/W2b`).
- **2000–2001 publication, fifteen Sumatera Barat districts.** BPS used a single uniform denominator (~62,000) for every district regardless of actual population (which ranges from 42,000 to 743,000). Corrected from BPS Query Builder Seri 2000 export (`correction_id = W2c`).

The same error likely exists in other provinces' per-capita publications, particularly 2017–2019; no other province has been cross-checked against BPS provincial websites.

7.7 Banten + West Java 2000–2001 (gap now filled from a supplementary BPS volume)

Banten was carved from Jawa Barat in 2000, and the main 2000–2001 publication's nominal tables did not cover Banten or post-split Jawa Barat (its constant-price base-1993 tables did, as did the separate per-capita publication), leaving ~28 nominal district-year gaps. These are filled from a supplementary BPS source, `PDRB_1998-2001_jabar_banten` (a nominal-only volume covering 1998–2001); the pipeline ingests its 2000–2001 tables, and Banten 2000–2001 is present in the `2000bounds` panel (24 nominal total-PDRB rows across the two years). Any residual absences reflect genuine source coverage limits; do not impute them.

The volume's 1998–1999 columns also cross-check the 1996–1999 parse; 38 of 52 comparable district-years agree within 5%. Details and the decision not to ingest the revised 1998–1999 values are in the research log.

7.8 Chained series: synthetic, non-additive levels

The chained panel backcasts real/2000 growth onto the real/2010 base. Backcast values (`is_backcast == True`) are synthetic 2010-price levels, not BPS-published, and are not additive across districts. Do not sum districts to obtain a province total from the chained panel. Use it for growth analysis only. Ten districts (`ratio_drift_flag == True`) show >5% disagreement between real/2000 and real/2010 growth at the 2012 validation year; treat their backcasts with extra caution.

The link ratio is taken at 2011; the overlap year 2012 was reserved for out-of-sample validation rather than averaging. The 2011 link also crosses not only a price-base change (2000 -> 2010) but the SNA 1993 -> 2008 methodology break, so pre/post-2011 level comparisons are less reliable than a pure rebase would suggest.

7.9 Pre-split aggregate rows double-count

BPS sometimes printed a child district's values before the formal split, marked #; these are flagged `is_pre_split_aggregate == True`. Including them in any district sum double-counts that area's output. Standard practice: `df[df.is_pre_split_aggregate == False]` before any aggregation.

7.10 Papua promulgation-date inconsistency in the split concordance

The UU 26/2002 Papua districts appear with two different promulgation dates in the concordance (25 October vs 11 December 2002). The year is correct (2002); only the exact date differs. Impact is limited to analyses that key on the exact law date.

7.11 Provisional rows extend to 2025

The 2021–2023 publication carries forward columns for 2024–2025 marked *angka sangat sementara* (very preliminary), flagged `is_provisional == True`. For any analysis not requiring 2024–2025, filter `df.year <= 2023`.

7.12 Residual year-misassignment risk for sparse split-off district rows

The parser places a lone value in the first year column when a newly-split district appears with data in only one or two columns; affected pages are re-parsed in vision mode to read column position from the rendered image. Name typos surface in the harmonizer's unmatched-name audit (`harmonize_unmatched.csv`); all cases are resolved via `name_corrections.csv`; the harmonizer reports zero unmatched rows. The residual risk is a typo that coincides exactly with a *different* valid district name, which would match silently to the wrong district. Analysts relying on exact split-year dynamics should verify those rows against the source PDFs; a pre-publication roster-completeness audit found no such collisions. Details in the research log.

7.13 Total-vs-per-capita coverage asymmetry

Eight districts appear in only one of the two published panels, reflecting differences in which BPS publications covered total vs. per-capita PDRB.

- **Three Jambi districts** (`JAMBI_BUNGO_TEBO`, `JAMBI_SAROLANGUN_BANGKO`, `JAMBI_TANJUNG_JABUNG`) appear in the total panel only: pre-split parent aggregates (each split c. 1999) present in the 1996–1999 total-PDRB source, for which no contemporaneous per-capita publication exists.
- **Five Papua Barat districts** (`PAPUA_BARAT_KAIMANA`, `PAPUA_BARAT_RAJA_AMPAT`, `PAPUA_BARAT_SORONG_SELATAN`, `PAPUA_BARAT_TELUK_BINTUNI`, `PAPUA_BARAT_TELUK_WONDAMA`) appear in the per-capita panel only: post-2000 splits from Sorong (all 2003), covered by per-capita publications but not by total-PDRB publications for the same years.

Analysts joining the two panels on district key should expect these 8 districts on one side only.

7.14 Per-capita panel is not territory-consistent across split years

The 2000-vintage aggregation applies to the total panel only: post-2000 child districts are summed back onto their parent, so a parent's total-PDRB series covers the same territory throughout. Per-capita values are not additive, so the capita panel cannot use the same mechanism; child districts are dropped, and a split parent's per-capita series continues as published by BPS, covering only the rump parent territory after the split.

A silent territory change occurs mid-series wherever a 2000-vintage parent split after 2000. The starkest case is SUMBAWA (NTB): Sumbawa Barat split off in 2003 taking the Batu Hijau copper-gold mine, and Sumbawa's per-capita series drops from 13.7M rupiah (2003, full territory) to 4.7M (2004, rump), a –66% step that is entirely boundary change, not economics. A cross-check of implied population ($\text{total} \div \text{per-capita}$) flags roughly a hundred such discontinuities, concentrated at pemekaran years; no order-of-magnitude discrepancies appear, and large disagreements fall exactly where boundary changes predict.

The capita `_2000bounds` panel is boundary-consistent for districts that never split, and boundary-consistent *within* the pre-split and post-split segments of districts that did. For analyses requiring a boundary-consistent per-capita series across a split year, either restrict to never-split districts, split the series at the pemekaran year, or construct per-capita as total-panel PDRB divided by an external population series on 2000-vintage boundaries. Population-weighted aggregation of child per-capita values would require a district-level population series on consistent boundaries.

7.15 Undetected errors

The correction pass (Section 9) targeted errors we identified during construction: OCR artifacts, confirmed wrong denominators, and BPS source errors with an external cross-check. Errors we did not identify — dropped digits, transposed values, wrong denominators in unchecked provinces — appear in the panel without a warning. The DAPOER cross-check (Section 8) provides the strongest external validation, but it covers only the series and years DAPOER holds and cannot catch errors consistent across both sources. For analyses sensitive to individual cell values, particularly for lightly-covered districts or years with a single source, verify against the source PDFs.

7.16 Nominal series: the 2011 SNA2008 break

`value_standardized` for `table_type == "nominal"` is spliced from nine BPS publications. Eight of the nine publication boundaries inside the panel are clean — the median district year-on-year nominal ratio sits at 1.08–1.16, consistent with real growth plus inflation. The exception is the 2008–2010 | 2011–2013 boundary: median 2010→2011 ratio 1.65, 84 of 347 districts above 2×, sum-of-districts nominal Rp 4,708 T → Rp 7,227 T (+53% in one year against actual national nominal growth of roughly +15%). This is BPS's shift from the 2000 base year under the 1993 SNA to the 2010 base year under the 2008 SNA — broader sector coverage and a benchmark revision — not economic activity. The 2011–2013 publication tabulates 2011 onward only (2010 is the price base, not a data year), so the panel contains no year measured on both vintages, and nominal levels either side of 2011 are not directly comparable.

Consequences: (a) do not compute nominal growth across 2010–2011; (b) analysis pooling pre- and post-2011 nominal levels in an unbalanced panel, or with district-specific time trends, absorbs the break; (c) a series built as nominal ÷ CPI inherits the jump. For continuous real magnitudes use the chained panel (Section 7.8), which links across this seam using the 2011–2012 overlap between the 2000-base and 2010-base constant-price series. A separate, milder elevation appears at the 2001→2002 boundary (median 1.21; 14 districts above 1.8×), likely a minor 1993-base-tail vintage effect and/or post-crisis volatility.

8. DAPOER Cross-Validation

We cross-validated the panel against the World Bank INDO-DAPOER⁵, which covers 2000–2020 at district level and includes both SNA 1993 real (base 2000) and SNA 2008 real (base 2010) series. The cross-check covers five series: SNA 1993 nominal and real, each including and excluding oil and gas, plus SNA 2008 real. We did not cross-check SNA 2008 nominal, because it shares the same

⁵World Bank Indonesia Database for Policy and Economic Research (DAPOER): <https://databank.worldbank.org/source/indonesia-database-for-policy-and-economic-research> (accessed 14 August 2026).

extraction pipeline as the validated series. DAPOER's uniform IDR Million convention makes $\times 10 / \times 100$ discrepancies identifiable as data errors rather than unit mismatches.

3,311 of 3,358 real/2010 district-year comparisons agree within 1%. Of the 47 that do not, the 15 largest ($|\text{diff}| > 20\%$) all trace to defects on the DAPOER side, and no panel value was changed.

Three structural patterns explain all major nominal discrepancies

These are methodological differences, not errors in either source:

1. **DAPOER backdates district splits (2000–2003).** For ~120 districts created 1999–2003, DAPOER applies current (post-split, smaller) boundaries retroactively using sub-district data, whereas we report the historical publication value (the larger pre-split district); the two agree exactly from the year each split first appears in BPS publications. This difference is by design: our panel is built on the 421 districts of the 2000-vintage administrative map; 361 appear in at least one source publication, the remaining 60 in none. Sources reporting more districts (e.g., DAPOER's 548) reflect current boundaries and therefore contain units that do not exist for most of the sample period.
2. **SNA 2008 publication vintage (2011–2013).** DAPOER's SNA 1993 nominal series ends at 2013 with lower values than our panel, which switches to SNA 2008 from 2011. This reflects different accounting methodologies, not a panel error.
3. **Riau/Kepri petroleum attribution (2000–2001).** DAPOER distributes petroleum revenue to post-split districts retroactively; we follow the historical BPS attribution.

real/2010 cross-check: 15 flagged discrepancies ($|\text{diff}| > 20\%$)

Beyond the structural patterns, the real/2010 cross-check flagged 15 residual discrepancies, all DAPOER-side; no panel value was changed. The strongest rest on DAPOER contradicting *itself* — Sorong Kab, Takalar, and Malinau. A second group carries a clean typo signature: Asahan 2013 and Banda Aceh 2016. The weakest cases are where our panel is smooth and DAPOER is not: Halmahera Barat 2020 and the Morowali anomaly, where the evidence is internal consistency, not independent verification against a third source.

Per-flag mechanisms:

Flagged rows	DAPOER defect	Panel evidence
Sorong Kab 2013–2016 (4), Takalar 2013–2015 (3)	×100 data-entry error	Panel series smooth and internally consistent; DAPOER/ panel ratio is 99.98–100.01 exactly. A ×100 factor would make these districts larger than Jakarta.
Malinau 2018–2019 (2)	End-window dropout / corrupt values	Panel 6,972/7,374/7,849/7,808 bn (2017–2020), smooth; DAPOER's 49/52 bn would require +155% single-year growth.
Asahan 2013 (1)	Dropped-digit entry error (not a unit error)	Panel 2011–2015: 16,940/ 17,872/18,893/20,003/21,117 bn, on a steady +5.5–5.9%/yr path, from the same publication as its clean neighbors. Ratio is 9.98, not 10.00: DAPOER's 1,893 is our 18,893 with the “8” dropped. No boundary/Inalum story needed.
Banda Aceh 2016 (1)	Dropped-digit entry error	Panel 12,725/13,480/13,937 bn (2015–2017), smooth; same ≈9.98 signature.
Halmahera Barat 2020 (1)	÷10 at end-of-series	Panel 2019–2021: 1,522/1,530/ 1,548 bn, seamless (confirmed across the 2020–2022 and 2021–2023 publications). DAPOER's 153 bn is exactly panel÷10 at the last year of its SNA-2008 series (dropped trailing digit / stale tail).
Morowali 2018–2019 (2)	Stale vintage	Panel 13,364/28,358/34,103/ 43,902 bn (2017–2020), the genuine nickel boom; DAPOER rejoins our value at 43,902 in 2020, so its own 2019/2020 step would be +155%, implausible given the district's otherwise smooth 2017–2020 trajectory.
Malang Kota 2013 (1); sub-20% tail 2014/2016/2017	Reversed block	Panel smooth; DAPOER's 2013 value is exactly our 2017 — the full 2013–2017 block is reversed end-for-end.

The real/2010 cross-check is complete: 3,311 of 3,358 comparisons agree within 1%, and all 15 residual flags are defects in DAPOER, not the panel. The 32 comparisons in the 1–20% band are also fully explained: three are the Malang Kota reversal tail; Kampar 2018–2019 and Bengkulu Kota

2015/2017 carry the same DAPOER block-defect signature; twenty-one fall in DAPOER’s provisional 2018–2020 window and reconverge by series end; four are isolated single-year noise. No panel value is suspect. Per-case detail is in the research log.

Circularity caveat. “DAPOER-side” attribution rests on evidence, not an independent source. We never verify DAPOER against a third database, so for those cases the evidence is “consistent with our panel, inconsistent with DAPOER.” The first two groups are stronger because they rest on DAPOER’s own internal contradictions.

Province-sum audit notes

Two separate province-sum audits were run. `validate_panel.py` (Stage 6, Check 4) sums district-level data rows within each province-year in the standardized files and compares them to the published province total, producing 21 flags in the final build. All are structural: sparse oil-excluded coverage (only 1–2 districts per province report tanpa-migas figures, so comparisons against the oil-excluded province total are uninformative), Irian Jaya 1996–1999 large negative gaps from the Fakfak nulling (§7.3), and a few provisional-coverage cases (Gorontalo, Bangka-Belitung, Aceh). In Kalimantan Timur, district sums exceed the province total by 5–10% — a genuine BPS accounting convention where district-level oil/gas attribution sums to more than the reconciled province headline.

A broader pre-deduplication audit (`scripts/validate_province_sums.py`, 2026-08-12) flags 842 conditions in the pre-dedup files; every flag traces to a structural artifact — dual-section row duplication, migas-reconciliation conventions, or sparse oil-excluded coverage — and the published panels are unaffected. Province-cluster breakdown is in the research log.

Overlap revisions. Check 5 confirmed that the 2020–2022 and 2021–2023 publications printed identical values for the 2021–2022 overlap years: all 2,094 district-year comparison records show `revision_pct` = 0.0. BPS did not revise any district figures between the two editions. The panel’s source-priority rule (§4, Stage 5) uses the 2021–2023 file at those years; because the values are identical, there is no vintage ambiguity at this seam.

DAPOER extract file reference (`dapoer_extract/`)

Two files, both exports from the World Bank’s Data Bank platform in its standard wide (one column per year) format, unmodified from the World Bank’s own download:

File	Rows	Columns	Content
<code>dee68668-..._Data.csv</code>	3,841	Provinces Name, Provinces Code, Series Name, Series Code, then one column per year (1976 [YR1976] ... 2020 [YR2020])	The indicator values used in the cross-check above: GDP (SNA 1993 and SNA 2008, current and constant price, with/without oil and gas) and total population, by province and year.
<code>dee68668-..._Series - Metadata.csv</code>	3,849	Same key columns, plus a <code>Topic</code> field	Indicator-level metadata for the codes in <code>Data.csv</code> — units, periodicity, and topic classification.

On the `Topic` field: the GDP indicators are classified `Economic Indicators`; the population indicator (`SP.POP.TOTL`) is classified `Social and Demographic Indicators`. That split is the World Bank’s own indicator taxonomy, unrelated to anything in our pipeline — population is simply catalogued separately from GDP. Both files have a small number (3)

of duplicate rows, not further diagnosed here since they are an external export this project does not generate.

9. Corrections and Provenance

Tracing any value to its source

Every observation carries `source_file` and `source_priority`. To see what a publication printed before harmonization aggregated child districts, read the corresponding `_native` panel, or the parsed CSV in `parsed_csvs/` (replication package). The parsed CSVs are verbatim transcriptions of the PDF pages and can be diffed directly against `raw/pdf/`. Corrections are applied in Stage 2 to produce `corrected_csvs/` (regenerated at runtime); the sidecar records exactly which cells were modified.

The correction sidecar

`scripts/apply_corrections.py` (Stage 2) applies the corrections and logs them in `corrections/corrections_applied.csv`. The sidecar records one row per source cell each correction touched:

Column	Definition
<code>source_file</code>	Which of the 21 parsed CSVs the corrected cell lives in.
<code>fix_id</code>	The correction family identifier (= <code>correction_id</code> on the panel; see “Major correction categories” below).
<code>region_name</code>	District name as it appeared in the source row being corrected.
<code>year</code>	Year of the corrected cell.
<code>table_type</code>	nominal or real.
<code>raw_text_hash</code>	Stable content hash identifying the exact parsed row the fix modified — the join key <code>build_panel.py</code> uses to attach <code>correction_id</code> to panel rows (below).

The sidecar does not store the before/after values themselves; those live in `CORRECTIONS_BY_FILE` in `apply_corrections.py`’s own source. To see a correction’s actual values, read the script or diff `parsed_csvs/` against `corrected_csvs/` (regenerated at runtime). The sidecar is the authoritative record of **what** was changed and **why**; **36 of its 166 rows are exact duplicates**, a byproduct of resumed pipeline runs (“Parsed CSV schema” in Section 3) — they log the same correction twice, not two different ones, so use distinct-row counts (below) for any exact reconciliation.

How `correction_id` reaches the panel. The `raw_text_hash` is assigned at parse time and survives into the panel-input files, so `build_panel.py` attaches `correction_id` by joining the sidecar on `(source_file, raw_text_hash)` rather than on district name and year, which would mistag sibling rows that share a name and year but were not corrected. A build-time gate verifies that tagging is exact; some fixes tag zero panel rows (values nulled or superseded by a later publication). Implementation details are in the research log.

`correction_id` is carried on the `_native` panels and the chained panel but not on the aggregated `_2000bounds` panels (Section 5); for those, use the sidecar directly. External source CSVs used for recoveries (W2c, W2/W2b) are in the full project repository (`pipeline_out/full_runs/correction_sources/`), not the replication package.

Major correction categories

Fix identifiers are stable labels; most follow a mnemonic: `w`-prefixed families correct wrong values (`w2*` for wrong population denominators, `w3*` for digit-level garbles), `p`-prefixed families correct parse- or page-level errors or recover values the parser missed, and the remainder carry descriptive names (`OCR_garble`, `OCR_col_shift`, `KALTARA_shift`, `FAKFAK_err`, `BUG_E`). The numbers and letter suffixes distinguish families only; the prefix does not indicate whether an error is BPS-side or pipeline-side; that attribution is stated in each family's description. Unless noted otherwise, a corrected value appears in the published panel as shown; **[dropped]** means the row was nulled and removed before the panel was built, and **[superseded]** means the value was corrected in the source file but a later, higher-priority publication's value is what actually appears in the panel.

- **P7: Kota Bogor, 2002–2006.** `P7_real` recovers the real series 2002–2004, consistent with the following publication (`PDRB_2005–2007`); `P7_recover` recovers the nominal series 2002–2004 from `PDRB_2002–2004.pdf` p. 71; `P7_garble` fixes nominal 2005–2006 (garbled to impossible values) **[superseded]**.
- **P6/P6_recover: Kab. Sumba Barat, 2005–2007.** Garbled OCR values for 2005–2006, nulled (`P6`) **[dropped]** the 2007 value initially nulled under the same fix was confirmed correct via the source PDF and recovered (`P6_recover`).
- **P_recover: Mandailing Natal, 1997.** OCR left both duplicate nominal rows blank; recovered from the BPS PDF.
- **W2c: Sumatera Barat per-capita 2000–2001.** BPS used a uniform wrong population denominator for all 15 districts; corrected from the BPS Query Builder Seri 2000 export.
- **W2/W2b: Lampung per-capita 2017–2019.** Five districts used wrong population denominators; corrected from the BPS Lampung provincial website (`w2b`). The same source file's 2020–2021 cells for Bandar Lampung and Metro were corrected too (`w2`) but are superseded by later publications **[superseded]**.
- **W3 series: digit garbles.** Banjarmasin 1997 real (scan-shadow artifact, `w3`); Kampar real 2000–2003 (BPS source error, roughly 160x too low, `w3a`) **[dropped]** Musi Rawas 2001 nominal oil-excluded (OCR garble, `w3b`); Sambas per-capita 1996–1999 (+10 denominator error, `w3c`); Sumenep 1996 real (dropped decimal period, `w3d`); Mamuju 1999 real (232% implausible spike, `w3e`) **[dropped]** Kapuas 1999 per-capita (spurious leading digit, `w3f`).
- **OCR garbles.** Eight additional corrections for stray characters producing non-parseable values.
- **OCR_col_shift: Luwu Utara, 1997–1999.** Parser aligned columns one position left; 1997 nulled, 1998–1999 shifted back to their correct year.
- **BPS source errors, confirmed and dropped.** Barru 1998 nominal/per-capita (`P5a`) and Mamuju 1999 real (`w3e`, also listed above under W3 series): BPS-side errors, not pipeline artifacts **[dropped]**.
- **KALTARA_shift: Kalimantan Utara per-capita 2011–2013, four districts (25 corrections).** The PDF parser merged BPS Tabel 160 (Kalimantan Utara) into the preceding Tabel 159 (Kalimantan Timur) and shifted the Kaltara rows one year-column left. Content-keyed corrections restore Bulungan, Malinau, Nunukan, and Tarakan per-capita values to their printed year columns (verified against the source PDF).
- **BUG_E: Riau per-capita, 2008–2012.** A mislabeled oil-exclusion flag on two districts (Kab. Indragiri Hulu, Kab. Pelalawan) caused duplicate rows; the spurious flag is cleared so `oil_excluded` is assigned correctly downstream.

144 corrections (22 fix families across 9 source files) produce 166 sidecar rows in `corrections_applied.csv`; corrections touching both nominal and real cells for the same district-year produce multiple rows, so 144 is the logical-fix count and 166 the cell count.

Scope of the correction pass

The 144 corrections address errors that were identified and verified during construction. The correction pass does not guarantee the absence of undetected errors: BPS source errors that were never flagged (dropped digits, transposed values, wrong denominators in provinces not cross-checked) will appear in the panel without a `correction_id`. For cell-sensitive analyses, verify in `parsed_csvs/` and, if needed, the source PDF.

Errata and contact

We welcome inquiries, extensions, corrections, and additional evidence, especially from users with access to BPS regional publications. Attributions of source anomalies (Fakfak 1996–1999, DAPOER-side classifications in [Section 8](#)) rest on available evidence. The project repository is at <https://github.com/nayfajo/indonesia-subnational-gdp>.

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Licensing terms are set on the PURR deposit record.

End of data guide.